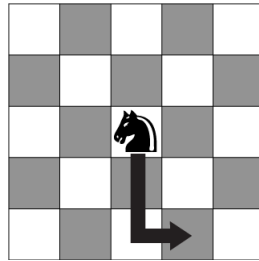


The Penguin's Tour

Aditya Pahuja

1 Introduction

The *knight's tour* is an ancient puzzle involving a rectangular grid and a knight. A knight is a chess piece that moves two squares horizontally or vertically on the grid and one square in the perpendicular direction, like this:



The goal of the knight's tour is to find a path that the knight can take such that it lands on every single square on the board exactly once. A *closed knight's tour* is a knight's tour that ends with the knight moving to the square that it started on, while an *open knight's tour* is a tour in which the knight does not end on its starting square.

2 The Puzzle

Let a *penguin* be an object that, analogous to a knight, can move three squares horizontally or vertically and two squares in the perpendicular direction. A *closed penguin's tour* would then be a path that a penguin can take such that it lands on every square of the grid exactly once and lands on its original square. Prove that a “closed penguin's tour” is impossible on A) a 7×7 grid and B) a 6×6 grid.

